

SafeCharge

Interactive BMS Digital Twin

Group 6

Digital Engineering | Otto-von-Guericke-University Magdeburg

Sprint 2 — Full Implementation Update

May 2026 | Live Unity demo running end-to-end

From Sprint 1 to Sprint 2

Sprint 1 — Delivered

- ✓ Concept & architecture locked
- ✓ Safety thresholds defined (45 / 60 °C)
- ✓ Unity + GitHub pipeline ready
- ✓ SOH gap identified from professor feedback

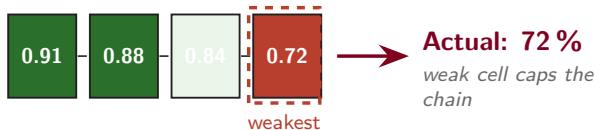


Sprint 2 — Now Live

- Working Unity simulation with real-time physics
- CC-CV charging profile + auto-stop at 100 %
- Per-cell SOH degradation modelled over cycles
- Thermal hysteresis band (42–45 °C)
- EV range: displayed vs. actual gap in km
- Premium dark UI with rounded panels

Professor's Feedback — What the Gauge Hides

Pack gauge reads 80 %



Professor's point

"If the gauge shows 80 %, that does not mean 80 % is usable. The weakest cell limits the whole chain. Show per-cell health *and* the real driving range consequence."

Sprint 2 response: (1) 4×4 cell SOH heatmap. (2) Dual SOC readout: *displayed* vs. *actual*. (3) EV range estimator shows km gap from cell ageing.

Sprint 2 — Feature Overview

Coulomb Counting SOC

Integration of current over time gives real-time SOC.

CC–CV Charging Profile

Constant current to 80 %, then tapers. Auto-stop at 99.9 %.

Joule Heating & Thermal

I^2R heat, lumped mass model, convection. Fan ON at 45 °C.

Hysteresis Thermostat

Fan ON above 45 °C, OFF only below 42 °C. No oscillation.

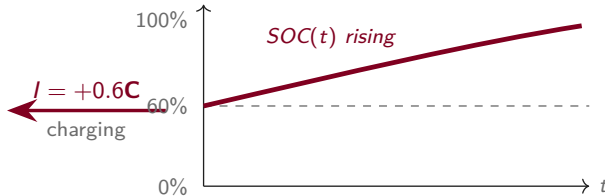
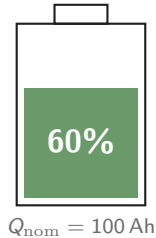
Cycle-Life Degradation

$\text{SOH}_i = 1 - k_i N$. Each cell fades at its own rate.

EV Range Estimator

Healthy vs. actual km gap shows the real cost of SOH loss.

State of Charge — Coulomb Counting



App convention

Slider right $\Rightarrow I > 0$
(charging, SOC rises)

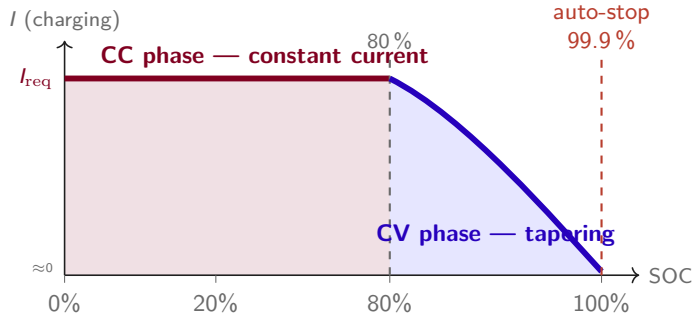
Slider left $\Rightarrow I < 0$
(discharging, SOC falls)

$$SOC(t) = SOC_0 + \frac{1}{Q_{\text{nom}}} \int_0^t I(\tau) d\tau$$

| $I > 0$: charging, $I < 0$: discharging

Implemented as $\Delta SOC = I \Delta t / Q_{\text{nom}}$ every physics step. Source: Plett (2015), BMS Vol. I.

CC–CV Charging Profile



CV taper formula

$$\alpha = \frac{1 - \text{SOC}}{1 - \text{SOC}_{\text{cv}}}$$

$$I_{\text{eff}} = I_{\text{req}} \cdot \text{clamp}(\alpha, 0, 1)$$

Parameters

$$\text{SOC}_{\text{cv}} = 0.80$$

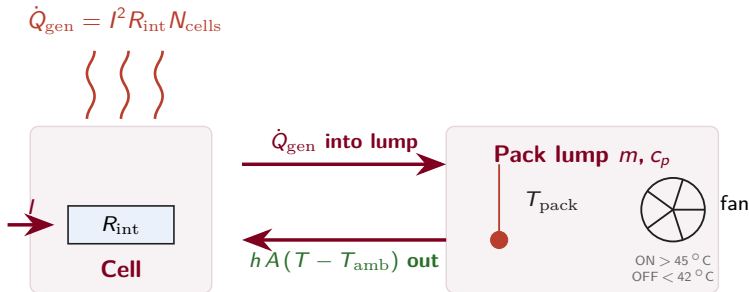
Stop when $\text{SOC} \geq 0.999$

At SOC = 93 %

$$\alpha = 0.07 / 0.20 = 0.35$$

Only 35 % of I_{req} flows

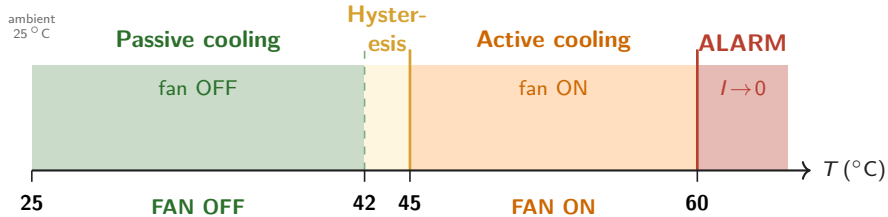
Joule Heating & Lumped Thermal Model



$$m c_p \frac{dT}{dt} = I^2 R_{\text{int}} N_{\text{cells}} - h A (T - T_{\text{amb}})$$

$h_{\text{passive}} = 10 \text{ W/m}^2 \text{K}$, $h_{\text{active}} = 40 \text{ W/m}^2 \text{K}$. Sources: Bernardi et al. (1985); Forgez et al. (2010).

Thermal Safety — Hysteresis Thermostat



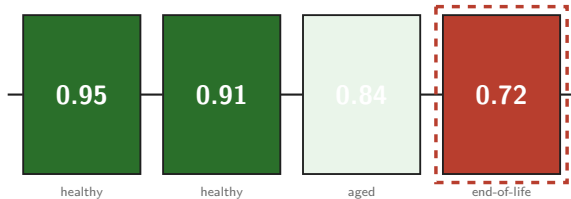
Why hysteresis?

Without the 3 °C band the fan would flicker ON/OFF every physics tick at exactly 45 °C, locking temperature. **With the band:** fan stays ON until 42 °C — smooth, stable control.

Alarm auto-clear

Alarm latches at 60 °C (current cut to zero). Auto-clears when T drops back below 45 °C so the user can resume charging after the pack has cooled.

SOH & the Weakest-Cell Rule



Pack ceiling = 0.72

Per-cell SOH

$$SOH_j = Q_{full,j} / Q_{nom}$$

Pack capacity (series)

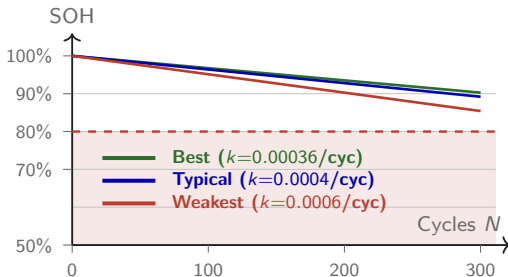
$$Q_{usable} = \min_j Q_{full,j}$$

Actual vs. displayed SOC

$$SOC_{actual} = SOC_{disp} \cdot SOH_{pack}$$

Cells in series: the weakest link sets the pack limit. Sources: Dubarry et al. (2009); IEC 62660-1:2018.

Cycle-Life Degradation Model



Degradation formula

$$\text{SOH}_i(N) = \max(0.5, 1 - k_i N)$$

Rate constants

$$k_{\text{base}} = 0.0004 \text{ /cycle}$$

$$k_i = k_{\text{base}} \pm 10\% \text{ spread}$$

$$k_{\text{weak}} = +0.0002 \text{ (1 cell)}$$

Cycle counting

$$N = \frac{\int |I| dt}{3600 Q_{\text{nom}}}$$

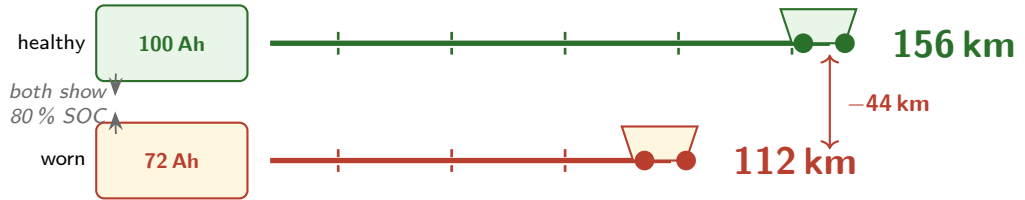
(Ah throughput based)

Key result

80 % SOH at ≈ 500 cycles

Floor: 50 % (minimum usable)

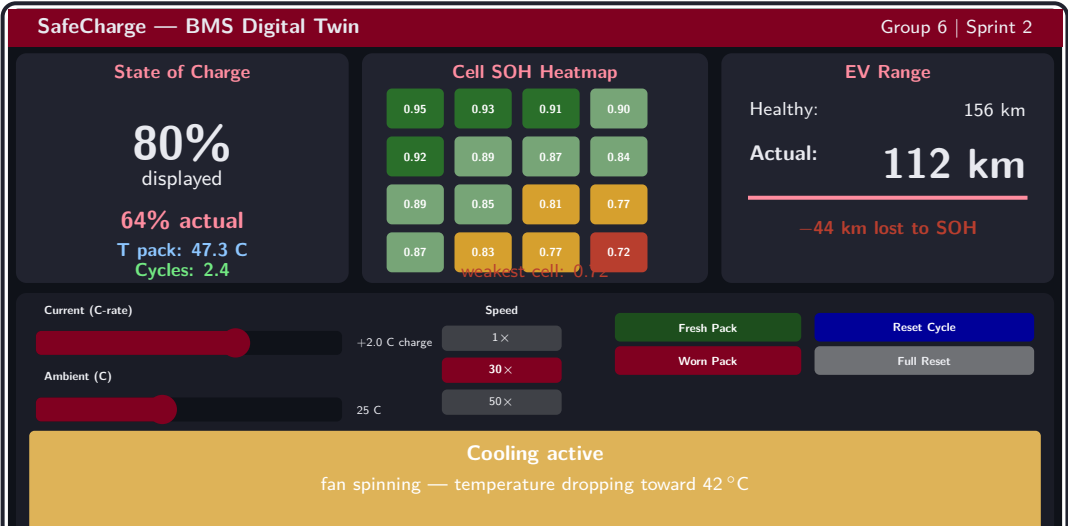
EV Range — Closing the Gap



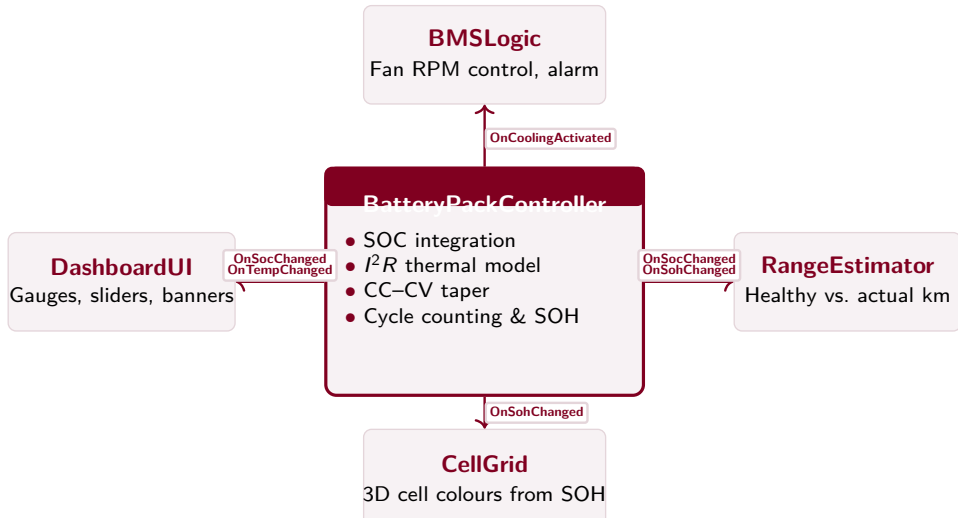
$$E_{\text{usable}} [\text{kWh}] = V_{\text{nom}} \cdot Q_{\text{usable}} \cdot \text{SOC}_{\text{disp}} / 1000 \quad | \quad \text{Range} [\text{km}] = E_{\text{usable}} / c_{\text{drive}}$$

$V_{\text{nom}} = 350 \text{ V}$, $Q_{\text{nom}} = 100 \text{ Ah}$, $c_{\text{drive}} = 0.18 \text{ kWh/km}$. Source: Larminie & Lowry (2012).

Dashboard UI — Premium Dark Theme



BMS Software Architecture — Event-Driven

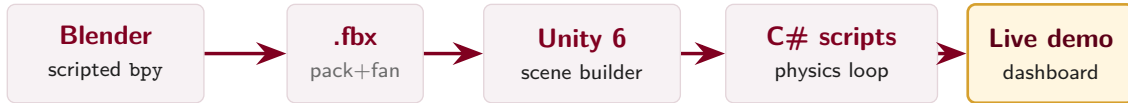


Model Parameters & Assumptions

Param.	Value	Meaning / Note	Param.	Value	Meaning / Note
Q_{nom}	100 Ah	Pack nominal capacity	T_{cool}	45 °C	Fan trigger threshold
V_{nom}	350 V	Nominal pack voltage	T_{hyst}	42 °C	Fan OFF (hysteresis)
R_{int}	0.030 Ω	Internal resist. per cell	T_{alarm}	60 °C	Alarm threshold
N_{cells}	16	Cells in series (4×4)	c_{drive}	0.18 kWh/km	EV energy per km
m	25 kg	Pack mass (approx.)	k_{base}	0.0004/cyc	Capacity fade/cycle
c_p	800 J/kgK	Specific heat (LiFePO ₄)	k_{weak}	+0.0002/cyc	Extra fade, 1 weak cell
A	0.50 m ²	Simplified surface area	SOC_{cv}	0.80	CV phase threshold
h_{pass}	10 W/m ² K	Passive convect. coeff.	SOC_{stop}	0.999	Auto-stop threshold
h_{act}	40 W/m ² K	Active (fan ON) coeff.	timeScale	30 ×	Demo speed multiplier

Note on surface area: $A = 0.50 \text{ m}^2$ is a placeholder. A realistic 16-cell prismatic module exposes $\approx 0.28\text{--}0.35 \text{ m}^2$, making passive cooling weaker and thermal rise more realistic. Reduce A in the Unity Inspector to tune.

3D Asset & Unity Pipeline



Blender (Python script)

- Pack shell + 4×4 prismatic cells
- Cells named cell_00...cell_15 (index = SOH array)
- Cooling fan: housing + 5-blade rotor
- Cell cross-section for ion-flow particles (Sprint 4)
- All geometry authored via bpy API

Unity C# Scripts

- **BatteryPackController**: SOC integration, thermal model, CC-CV, SOH degradation, cycle counting
- **BMSLogic**: fan RPM control, alarm at 60 °C, thresholds at 45/42 °C
- **CellGrid**: drives 3D cell colours directly from the SOH array
- **RangeEstimator**: healthy vs. actual km gap from usable capacity
- **DashboardUI**: all gauges, sliders, speed buttons, banners, reset buttons

Roadmap — Sprint 3 & Beyond



Sprint 2

Done
May 2026

Sprint 3

- Tutorial overlay
- SOC ring animation
- Scenario presets

Jun 2026

Sprint 4

- Ion-flow particles
- Per-cell thermal coupling
- Result summary card

Jun 2026

Sprint 3 focus: make the app self-explanatory for a non-engineer audience — guided mode, hover tooltips, scenario presets (new pack / 2-year worn / near EOL). The physics engine is complete; effort shifts to UX polish.

Thank You

We are open for questions.

Group 6 | Sprint 2 | May 2026

Live Unity demo available — ask us to run any scenario.